



I2C – BH17 테스트

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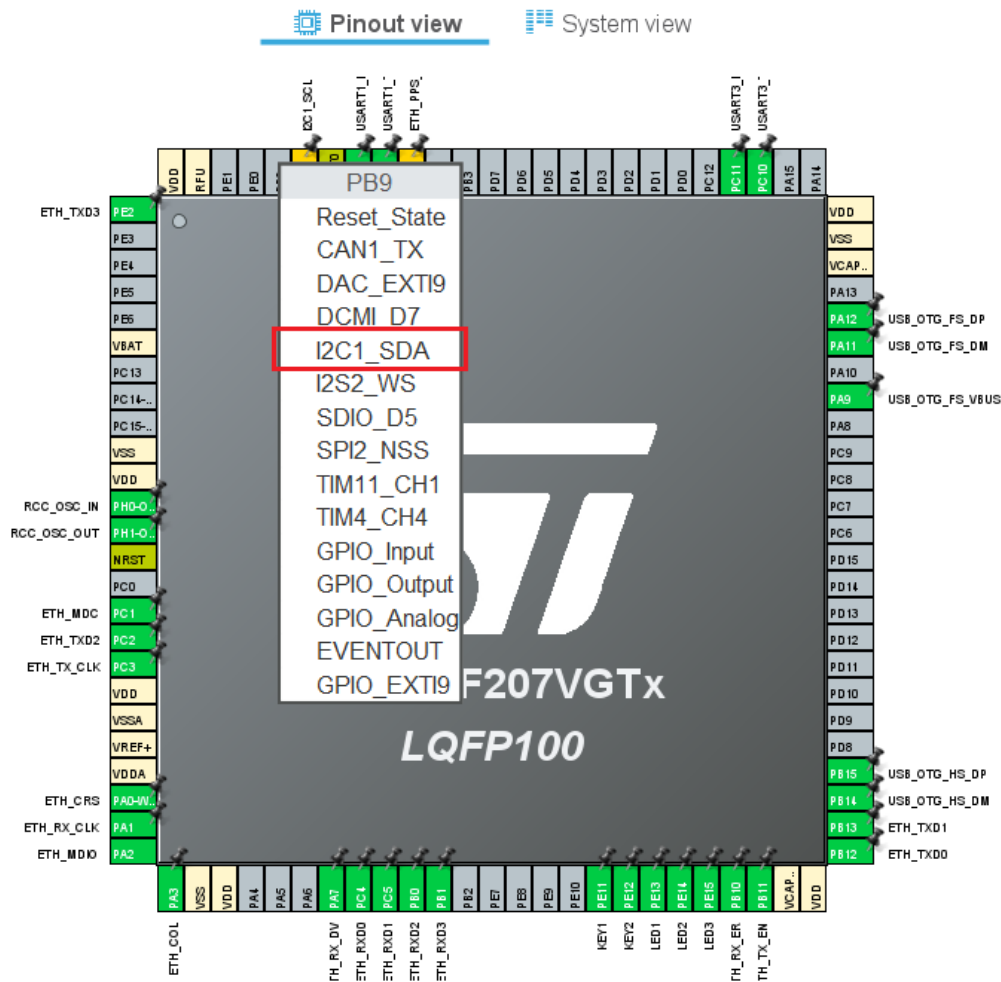
Mango-M32F2 I2C-BH17 조도 sensor 테스트

I2C pinmap

I2C – PB8, PB9

PB7	93	PB7	UART1_TX
PB8	95	PB8	I2C1_SCL
PB9	96	PB9	I2C1_SDA
PB10	47	PB10	MII_RX_FR

- PB8-I2C1_SCL, PB9-I2C1_SDA 선택.



I2C 설정

- “Connectivity” 클릭, “I2C1” 클릭.
- “Mode” - “I2C” 클릭.

The screenshot displays the 'bma280_prj.ioc - Pinout & Configuration' window. The 'Pinout & Configuration' tab is active, showing a list of components on the left and configuration options on the right. In the left sidebar, under the 'Connectivity' category, 'I2C1' is selected and highlighted with a red box. The main panel shows the 'I2C1 Mode and Configuration' settings. The 'Mode' dropdown menu is set to 'I2C' and is also highlighted with a red box. Below this, the 'Configuration' section includes a 'Reset Configuration' button and tabs for 'NVIC Settings', 'DMA Settings', 'GPIO Settings', 'Parameter Settings', and 'User Constants'. The 'Parameter Settings' tab is active, showing a search bar and a list of parameters. The 'Master Features' section is expanded, showing 'I2C Speed Mode' set to 'Standard Mode' and 'I2C Clock Speed (Hz)' set to '100000'. The 'Slave Features' section is also expanded, showing 'Clock No Stretch Mode' set to 'Disabled' and 'Primary Address Length' set to '7 bit'.

BH17 Test Project

■ Code 생성 후 main.c 수정

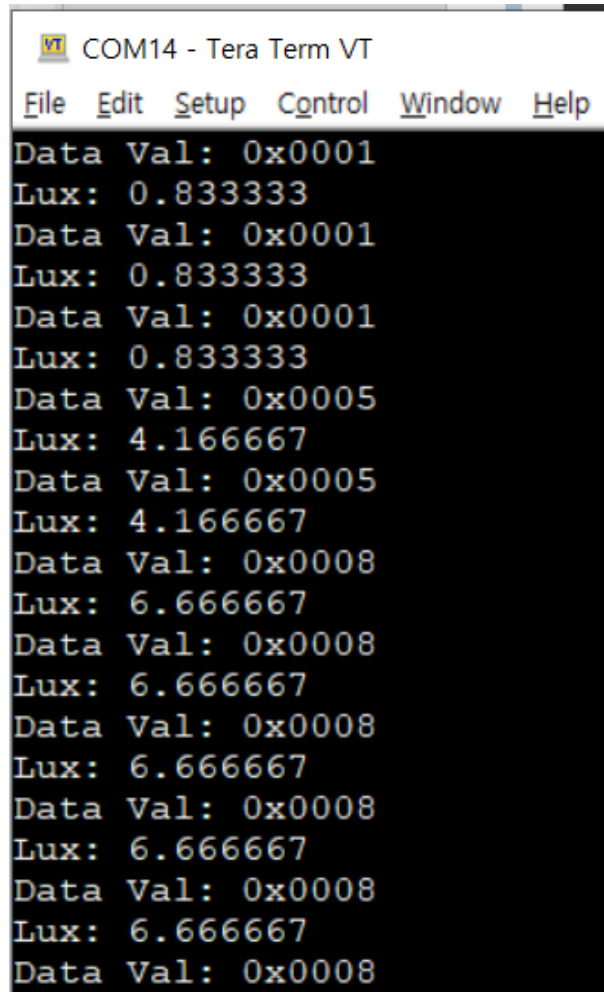
```
36  /* Private define ----- */
37  /* USER CODE BEGIN PD */
38
39  #define AL_BH17_SLAVE_ADDRESS (0x23 << 1)
40
41  /* USER CODE END PD */
42
43  static uint8_t SENSOR_AL_BH17_Read_Bytes(uint8_t devAddr, uint8_t length, uint8_t *data)
44  {
45      HAL_StatusTypeDef i2cRW_status;
46
47      i2cRW_status = HAL_I2C_Master_Receive(&hi2c1, devAddr, data, length, 1000);
48      if(HAL_OK != i2cRW_status)
49      {
50          printf("Read error:%d\r\n", i2cRW_status);
51          return 0;
52      }
53
54      return length;
55  }
56
57  static int8_t SENSOR_AL_BH17_Write_Bytes(uint8_t devAddr, uint8_t length, uint8_t *data)
58  {
59      HAL_StatusTypeDef i2cRW_status;
60
61      i2cRW_status = HAL_I2C_Master_Transmit(&hi2c1, devAddr, data, length, 1000);
62      if(HAL_OK != i2cRW_status)
63      {
64          printf("Write error:%d\r\n", i2cRW_status);
65          return -1;
66      }
67
68      return 0;
69  }
70  /* USER CODE END 0 */
```

BH17 Test Project

```
136 int main(void)
137 {
138     /* USER CODE BEGIN 1 */
139     uint8_t dataVal[2];
140     uint16_t data16Val;
141     float alData;
142     /* USER CODE END 1 */
143
144
145
146
147
148
149     /* USER CODE BEGIN 2 */
150     printf("I2C-BH17 test\r\n");
151     httpd_init();
152
153     HAL_Delay(5000);
154     dataVal[0] = 0x10;
155     SENSOR_AL_BH17_Write_Bytes(AL_BH17_SLAVE_ADDRESS, 1, dataVal);
156     HAL_Delay(100);
157     /* USER CODE END 2 */
158
159     /* Infinite loop */
160     /* USER CODE BEGIN WHILE */
161     while (1)
162     {
163         MX_LWIP_Process();
164
165         if(count++ % 10000 == 0)
166         {
167             SENSOR_AL_BH17_Read_Bytes(AL_BH17_SLAVE_ADDRESS, 2, dataVal);
168             data16Val = (dataVal[0] << 8) | dataVal[1];
169             printf("Data Val: 0x%04X\r\n", data16Val);
170             alData = (float)data16Val / 1.2;
171             printf("Lux: %f\r\n", alData);
172         }
173     }
```

Test 결과 확인

- CS-BH17 센서 보드를 8Pin Cable을 이용 CON7에 연결함.
- 손으로 센서를 가리면 Lux 값이 변함.



```
COM14 - Tera Term VT
File Edit Setup Control Window Help
Data Val: 0x0001
Lux: 0.833333
Data Val: 0x0001
Lux: 0.833333
Data Val: 0x0001
Lux: 0.833333
Data Val: 0x0005
Lux: 4.166667
Data Val: 0x0005
Lux: 4.166667
Data Val: 0x0008
Lux: 6.666667
Data Val: 0x0008
Lux: 6.666667
Data Val: 0x0008
Lux: 6.666667
Data Val: 0x0008
Lux: 6.666667
Data Val: 0x0008
Lux: 6.666667
Data Val: 0x0008
```


Thank You

